# Year II — Mistake & Methods Log

Based on lectures by Various Lecturers

Notes taken by Anonymous

Michaelmas - Epiphany 2026-2027

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# Chapter 1

# Mathematical Methods II

## 1.1 Line Integrals

### 1.1.1 Mistakes and Tips

**Scalar Field vs Vector Field**

A scalar field assigns a scalar value to each point in space, while a vector field assigns a vector to each point. For example, temperature can be modelled by a scalar field, whereas fluid velocity can be modelled by a vector field.

**Scalar Line Integral vs Vector Line Integral**

A scalar line integral accumulates a scalar field along a curve, weighted by arc length:

$$\int_C f \, d\ell.$$

If $\mathbf{x}(t)$ parametrises the curve for $t \in [a, b]$, then

$$\int_C f \, d\ell = \int_a^b f(\mathbf{x}(t)) \left| \frac{d\mathbf{x}}{dt} \right| dt.$$

For example, if $f$ is linear mass density, this integral gives the mass of a wire.

A vector line integral (also called a work integral or circulation integral) measures the component of a vector field along the direction of travel:

$$\int_C \mathbf{F} \cdot \hat{\mathbf{t}} \, d\ell = \int_a^b \mathbf{F}(\mathbf{x}(t)) \cdot \frac{d\mathbf{x}}{dt} \, dt.$$

Unlike a scalar line integral, a vector line integral depends on the orientation of the curve: reversing the direction reverses its sign.

**What does $f(\mathbf{x}(t))$ mean?**

The parametrisation $\mathbf{x}(t) = (x(t), y(t), z(t))$ gives the point on the curve at parameter value $t$. To evaluate a scalar field $f(x, y, z)$ along the curve, substitute $x = x(t)$, $y = y(t)$ and $z = z(t)$, giving $f(\mathbf{x}(t)) = f(x(t), y(t), z(t))$. For a vector field, substitute in the same way into each component.

**The speed $|\mathbf{x}'(t)|$ may vary**

The quantity

$$|\mathbf{x}'(t)| = \sqrt{x'(t)^2 + y'(t)^2 + z'(t)^2}$$

is the speed along the parametrised curve. It is generally a function of $t$, not a constant.

**Parametrising implicit curves**

If a curve is not already parametrised, choose a parameter that makes its equations and constraints easy to express. For circular or cylindrical symmetry, consider polar or cylindrical coordinates. Sketching the curve often helps identify the parameter range.

**Splitting piecewise-smooth curves**

If a curve consists of multiple smooth pieces, evaluate the integral on each piece and add the results. For vector line integrals, make sure each piece is parametrised in the direction of travel.

**Unit tangent vectors and orthonormal bases**

A unit tangent vector has magnitude 1. Unit tangent vectors associated with different coordinate directions are not automatically mutually orthogonal. Check their dot products before treating them as an orthonormal basis.

**Length of a curve**

If the question asks for the length of a curve, use the arc-length formula

$$L = \int_a^b |\mathbf{x}'(t)|\, dt = \int_{t_0}^{t_1} \sqrt{\frac{d\mathbf{x}}{dt} \cdot \frac{d\mathbf{x}}{dt}}\, dt.$$

**Orientation matters for vector line integrals**

Reversing the direction of travel changes the sign of $\int_C \mathbf{F} \cdot d\mathbf{x}$, but does not change a scalar line integral $\int_C f\, d\ell$.

### 1.1.2 Methods

**Method 1.1.1** (Evaluating a Scalar Line Integral)**.**

(i) Parametrise the curve as $\mathbf{x}(t) = (x(t), y(t), z(t))$, for $a \le t \le b$.

(ii) Differentiate to find $\mathbf{x}'(t)$.

(iii) Calculate the speed $|\mathbf{x}'(t)|$.

(iv) Substitute the parametrisation into the field to obtain $f(\mathbf{x}(t))$.

(v) Evaluate

$$\int_C f\, d\ell = \int_a^b f(\mathbf{x}(t))|\mathbf{x}'(t)|\, dt.$$

**Method 1.1.2** (Evaluating Arc Length).

(i) Parametrise the curve as $\mathbf{x}(t)$, for $a \leq t \leq b$.

(ii) Differentiate to find $\mathbf{x}'(t)$.

(iii) Calculate $|\mathbf{x}'(t)| = \sqrt{\mathbf{x}'(t) \cdot \mathbf{x}'(t)}$.

(iv) Integrate

$$L = \int_a^b |\mathbf{x}'(t)|\, dt.$$

**Method 1.1.3** (Evaluating a Vector Line Integral).

(i) Parametrise the curve in the required direction: $\mathbf{x}(t)$, $a \leq t \leq b$.

(ii) Substitute into the vector field to obtain $\mathbf{F}(\mathbf{x}(t))$.

(iii) Differentiate the curve to find $\mathbf{x}'(t)$.

(iv) Calculate the dot product $\mathbf{F}(\mathbf{x}(t)) \cdot \mathbf{x}'(t)$.

(v) Evaluate

$$\int_C \mathbf{F} \cdot d\mathbf{x} = \int_a^b \mathbf{F}(\mathbf{x}(t)) \cdot \mathbf{x}'(t)\, dt.$$

Do not multiply by $|\mathbf{x}'(t)|$ in this formula: the derivative vector already accounts for direction and parameter speed.

**Method 1.1.4** (Finding Unit Tangent Vectors in Orthogonal Coordinates). For a coordinate map $\mathbf{x}(u, v, w)$, define the scale factors

$$h_u = \left|\frac{\partial \mathbf{x}}{\partial u}\right|, \quad h_v = \left|\frac{\partial \mathbf{x}}{\partial v}\right|, \quad h_w = \left|\frac{\partial \mathbf{x}}{\partial w}\right|.$$

The corresponding unit coordinate tangent vectors are

$$\mathbf{e}_u = \frac{1}{h_u}\frac{\partial \mathbf{x}}{\partial u}, \quad \mathbf{e}_v = \frac{1}{h_v}\frac{\partial \mathbf{x}}{\partial v}, \quad \mathbf{e}_w = \frac{1}{h_w}\frac{\partial \mathbf{x}}{\partial w}.$$

These formulae apply where the scale factors are nonzero. The coordinate directions are orthogonal only if their tangent vectors are mutually perpendicular.

**Method 1.1.5** (Checking an Orthonormal Basis). Check that each vector has unit magnitude and that distinct vectors are mutually orthogonal:

$$|\mathbf{e}_i| = 1, \qquad \mathbf{e}_i \cdot \mathbf{e}_j = 0 \quad (i \neq j).$$

If these conditions hold, the vectors form an orthonormal set; in three dimensions, three such vectors form an orthonormal basis.

**Method 1.1.6** (Finding Vector Components in an Orthogonal Coordinate System).

(i) Rewrite the Cartesian components and the position in the desired coordinates using the coordinate transformations.

(ii) Find the unit coordinate vectors $\mathbf{e}_u, \mathbf{e}_v, \mathbf{e}_w$.

(iii) Express the field in the orthonormal basis by taking dot products:

$$F_u = \mathbf{F} \cdot \mathbf{e}_u, \qquad F_v = \mathbf{F} \cdot \mathbf{e}_v, \qquad F_w = \mathbf{F} \cdot \mathbf{e}_w.$$

(iv) Write $\mathbf{F} = F_u \mathbf{e}_u + F_v \mathbf{e}_v + F_w \mathbf{e}_w$.

Remember that changing the coordinate expressions of $x, y, z$ is not enough by itself: vector components must be resolved along the new unit coordinate vectors.

## 1.2 Surface and Volume Integrals

### 1.2.1 Mistakes and Tips

**Scalar Surface Integral vs Vector Surface Integral**

A scalar surface integral accumulates a scalar quantity over a surface. For example, if $f$ represents surface mass density, then

$$\iint_S f \, dS$$

gives the total mass distributed over the surface.

A vector surface integral is often used to calculate the flux of a vector field through an oriented surface:

$$\iint_S \mathbf{F} \cdot \hat{\mathbf{n}} \, dS.$$

The dot product with the unit normal measures the component of the field passing through the surface, rather than the component running parallel to it.

**Scalar Volume Integral vs Vector Volume Integral**

A scalar volume integral accumulates a scalar quantity throughout a three-dimensional region. For example, if $f$ is a mass density, then

$$\iiint_V f \, dV$$

gives the total mass within the volume.

A vector volume integral integrates each component of a vector field separately. Its result is a vector, not a scalar:

$$\iiint_V \mathbf{F} \, dV = \begin{pmatrix} \iiint_V F_x \, dV \\ \iiint_V F_y \, dV \\ \iiint_V F_z \, dV \end{pmatrix}.$$

This is not flux through the boundary; flux is a surface integral.

**Surface Area vs Scalar Surface Integral**

Surface area is the special case of a scalar surface integral where the integrand is 1:

$$\text{Area}(S) = \iint_S 1 \, dS.$$

If $\mathbf{x}(u, v)$ parametrises $S$, then

$$\text{Area}(S) = \iint_U \left| \frac{\partial \mathbf{x}}{\partial u} \times \frac{\partial \mathbf{x}}{\partial v} \right| du \, dv.$$

For a general scalar field $f$, use

$$\iint_S f\,dS = \iint_U f(\mathbf{x}(u,v)) \left|\frac{\partial \mathbf{x}}{\partial u} \times \frac{\partial \mathbf{x}}{\partial v}\right| du\,dv.$$

The notation Area($S$) should only be used when the integrand is 1.

**Volume vs Scalar Volume Integral**

The volume of a region is the special case of a scalar volume integral where the integrand is 1:

$$\text{Vol}(V) = \iiint_V 1\,dV.$$

Under a coordinate transformation $\mathbf{x} = \mathbf{x}(u,v,w)$,

$$\text{Vol}(V) = \iiint_{V'} |J|\,du\,dv\,dw,$$

whereas a general scalar volume integral is

$$\iiint_V f(\mathbf{x})\,dV = \iiint_{V'} f(\mathbf{x}(u,v,w))|J|\,du\,dv\,dw.$$

In Cartesian coordinates, $|J| = 1$. The integrand is still $f(x,y,z)$ for a general scalar volume integral; use 1 only when calculating the volume itself.

**Conceptual Understanding of Flux**

Flux measures the net amount of a vector field passing through an oriented surface:

$$\iint_S \mathbf{F} \cdot \hat{\mathbf{n}}\,dS.$$

The dot product selects the component of $\mathbf{F}$ in the normal direction. A field tangent to the surface contributes zero flux at that point. Reversing the normal reverses the sign of the flux.

**The 3D Jacobian**

For a coordinate transformation $\mathbf{x} = \mathbf{x}(u,v,w)$,

$$J = \frac{\partial(x,y,z)}{\partial(u,v,w)} = \det\begin{pmatrix} x_u & x_v & x_w \\ y_u & y_v & y_w \\ z_u & z_v & z_w \end{pmatrix} = \mathbf{x}_u \cdot (\mathbf{x}_v \times \mathbf{x}_w).$$

The absolute value $|J|$ is the local volume-scaling factor:

$$dV = |J|\,du\,dv\,dw.$$

In Cartesian coordinates the Jacobian determinant is 1. In a general transformation it need not be 1.

**Do not always parametrise the whole surface at once**

A surface such as a cube is piecewise smooth. Split it into its faces, calculate the contribution from each face, and add the results. For flux, ensure each face's normal has the required orientation.

**Distinguish the normal vector from its magnitude**

For a parametrised surface, $\mathbf{x}_u \times \mathbf{x}_v$ is a normal area vector. Its magnitude gives the surface-area scaling, while its direction gives the orientation. Do not replace this vector by its magnitude in a flux integral.

**Number of parameters**

A surface is generally parametrised by two parameters, while a three-dimensional volume is generally described using three coordinates or parameters.

**Choose coordinates to simplify the region**

Use Cartesian coordinates for regions bounded simply by planes or coordinate-aligned surfaces; cylindrical coordinates for circular symmetry or expressions involving $x^2+y^2$; and spherical coordinates for spheres and radial symmetry. Always sketch the region and check the bounds after changing coordinates.

**Standard Volume Elements**

- **Cartesian:**
$$dV = dx\,dy\,dz.$$
- **Cylindrical:** $x = r\cos\theta$, $y = r\sin\theta$,
$$dV = r\,dr\,d\theta\,dz.$$
- **Spherical:** using $\theta$ for the angle from the positive $z$-axis and $\phi$ for azimuth,
$$dV = r^2\sin\theta\,dr\,d\theta\,d\phi.$$

Check your lecture's spherical-coordinate convention, since some courses interchange $\theta$ and $\phi$.

### 1.2.2 Methods

**Method 1.2.1** (Systematically Finding Limits for a Volume Integral)**.**

(i) Sketch the region and identify its bounding surfaces.

(ii) Choose an order of integration, for example $dz\,dy\,dx$.

(iii) Find the lower and upper bounds for the innermost variable.

(iv) Project the region onto the plane of the two remaining variables and determine their bounds.

(v) Translate the bounds into the chosen coordinate system.

(vi) If one set of bounds cannot describe the region, split it into simpler regions and add the integrals.

**Method 1.2.2** (Evaluating Surface Area)**.**

(i) Parametrise the surface as $\mathbf{x}(u, v)$ over a parameter domain $U$.

(ii) Calculate $\mathbf{x}_u$ and $\mathbf{x}_v$.

(iii) Calculate the magnitude $|\mathbf{x}_u \times \mathbf{x}_v|$.

(iv) Evaluate

$$\text{Area}(S) = \iint_U |\mathbf{x}_u \times \mathbf{x}_v| \, du \, dv.$$

(v) For a piecewise-smooth surface, calculate the area of each piece and add them.

**Method 1.2.3** (Evaluating a Scalar Surface Integral)**.**

(i) Sketch or identify the surface and its parameter domain.

(ii) Parametrise the surface as $\mathbf{x}(u, v)$, or split a piecewise-smooth surface into faces.

(iii) Calculate $\mathbf{x}_u$, $\mathbf{x}_v$ and $|\mathbf{x}_u \times \mathbf{x}_v|$.

(iv) Substitute the parametrisation into the scalar field and evaluate

$$\iint_S f \, dS = \iint_U f(\mathbf{x}(u, v)) |\mathbf{x}_u \times \mathbf{x}_v| \, du \, dv.$$

(v) If the surface has multiple pieces, add the integral over each piece.

**Method 1.2.4** (Evaluating Flux Through a Parametrised Surface)**.**

(i) Parametrise the surface as $\mathbf{x}(u, v)$ over $U$.

(ii) Calculate $\mathbf{x}_u \times \mathbf{x}_v$.

(iii) Check the direction of this normal area vector. If it points opposite to the required orientation, reverse its sign.

(iv) Substitute $\mathbf{x}(u, v)$ into the vector field $\mathbf{F}$.

(v) Take the dot product with the *vector* normal area element and integrate:

$$\iint_S \mathbf{F} \cdot \hat{\mathbf{n}} \, dS = \iint_U \mathbf{F}(\mathbf{x}(u, v)) \cdot (\mathbf{x}_u \times \mathbf{x}_v) \, du \, dv.$$

(vi) For a piecewise-smooth surface, sum the flux through all pieces using the required orientation on each.

**Method 1.2.5** (Surface Given as a Graph $z = g(x, y)$)**.** For the upward orientation, parametrise the graph by $\mathbf{x}(x, y) = (x, y, g(x, y))$. Then

$$\mathbf{x}_x \times \mathbf{x}_y = (-g_x, -g_y, 1).$$

Hence, for a scalar field,

$$\iint_S f \, dS = \iint_D f(x, y, g(x, y)) \sqrt{1 + g_x^2 + g_y^2} \, dx \, dy,$$

and for flux,

$$\iint_S \mathbf{F} \cdot \hat{\mathbf{n}}\, dS = \iint_D \mathbf{F}(x, y, g(x, y)) \cdot (-g_x, -g_y, 1)\, dx\, dy.$$

For downward orientation, reverse the sign of the normal vector in the flux formula.

**Method 1.2.6** (Finding the Volume of a Region)**.**

(i) Sketch the region and determine the integration limits.

(ii) Choose Cartesian, cylindrical, spherical, or another suitable coordinate system.

(iii) Use the corresponding volume element, including its Jacobian:

$$\mathrm{Vol}(V) = \iiint_{V'} |J|\, du\, dv\, dw.$$

(iv) Integrate over the transformed region $V'$.

**Method 1.2.7** (Evaluating a Scalar Volume Integral)**.**

(i) Sketch the region and determine the limits of integration.

(ii) Choose coordinates that simplify the integrand or the region.

(iii) Rewrite the field in the new coordinates and calculate the absolute Jacobian $|J|$.

(iv) Evaluate

$$\iiint_V f(x, y, z)\, dV = \iiint_{V'} f(\mathbf{x}(u, v, w))|J|\, du\, dv\, dw.$$

(v) In Cartesian coordinates, $|J| = 1$; do not replace $f$ by 1 unless you are calculating the volume itself.

**Method 1.2.8** (Evaluating a Vector Volume Integral)**.** If $\mathbf{F} = (F_x, F_y, F_z)$, integrate each component separately:

$$\iiint_V \mathbf{F}\, dV = \begin{pmatrix} \iiint_V F_x\, dV \\ \iiint_V F_y\, dV \\ \iiint_V F_z\, dV \end{pmatrix}.$$

Use the same region, limits and volume element for all three components. Under a coordinate transformation, include the absolute Jacobian $|J|$ in each component integral.

## 1.3 Differential Operators

### 1.3.1 Mistakes and Tips

**Distinguishing scalar and vector fields**

A scalar field $f : \mathbb{R}^3 \to \mathbb{R}$ assigns a scalar to each point, whereas a vector field $\boldsymbol{f} : \mathbb{R}^3 \to \mathbb{R}^3$ assigns a vector.

- $\nabla f$ is a vector field.
- $\nabla \cdot \boldsymbol{f}$ is a scalar field.

– $\nabla \times \boldsymbol{f}$ is a vector field.

**Understanding the operator $\nabla$**

The symbol $\nabla$ is a differential operator, not an ordinary vector.

In Cartesian coordinates,

$$\nabla = \boldsymbol{e}_1 \frac{\partial}{\partial x} + \boldsymbol{e}_2 \frac{\partial}{\partial y} + \boldsymbol{e}_3 \frac{\partial}{\partial z}.$$

The expression $\boldsymbol{f} \cdot \nabla$ is an operator that differentiates in the direction of $\boldsymbol{f}$:

$$(\boldsymbol{f} \cdot \nabla) = f_1 \frac{\partial}{\partial x} + f_2 \frac{\partial}{\partial y} + f_3 \frac{\partial}{\partial z}.$$

When applied to a scalar field, it gives a directional derivative. When applied to a vector field, it differentiates each component.

**Gradient is not the same as directional differentiation**

The gradient $\nabla f$ points in the direction of steepest increase of $f$, and its magnitude is the maximum directional derivative.

For a unit vector $\boldsymbol{u}$, the directional derivative is

$$D_{\boldsymbol{u}} f = \boldsymbol{u} \cdot \nabla f.$$

Do not confuse the vector $\nabla f$ with the scalar $\boldsymbol{u} \cdot \nabla f$.

**Remembering the meaning of divergence**

Divergence measures the local net outward flux per unit volume.

$$\nabla \cdot \boldsymbol{f} > 0$$

indicates local net outflow, whereas

$$\nabla \cdot \boldsymbol{f} < 0$$

indicates local net inflow.

A zero divergence means there is no local net outflow, not necessarily that the vector field is zero.

**Remembering the meaning of curl**

Curl measures local circulation and the axis and direction of rotation.

In Cartesian coordinates, the first component is

$$(\nabla \times \boldsymbol{f})_1 = \frac{\partial f_3}{\partial y} - \frac{\partial f_2}{\partial z}.$$

Use the right-hand rule to interpret the direction of curl. A zero curl does not necessarily mean that the vector field itself is zero.

**Curl determinant signs**

When expanding the determinant for curl, the middle component has a minus sign in the cofactor

expansion.

$$\nabla \times \boldsymbol{f} = \begin{pmatrix} \partial_y f_3 - \partial_z f_2 \\ \partial_z f_1 - \partial_x f_3 \\ \partial_x f_2 - \partial_y f_1 \end{pmatrix}.$$

Always check the cyclic order of the components rather than memorising the formula without its signs.

### Using mixed partial derivatives

When proving

$$\nabla \times (\nabla f) = \mathbf{0} \quad \text{or} \quad \nabla \cdot (\nabla \times \boldsymbol{f}) = 0,$$

expand the components and use equality of mixed partial derivatives, assuming the required second derivatives are continuous.

For example,

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}.$$

### Applying product rules

Do not treat $\nabla$ as an ordinary vector when differentiating products. Derivatives act on every factor that depends on position.

For example,

$$\nabla \times (g\boldsymbol{f}) = (\nabla g) \times \boldsymbol{f} + g(\nabla \times \boldsymbol{f}).$$

The order of the cross product matters:

$$(\nabla g) \times \boldsymbol{f} = -\boldsymbol{f} \times (\nabla g).$$

### Distinguishing $\boldsymbol{f} \cdot \nabla$ from $\nabla \cdot \boldsymbol{f}$

These expressions are different operators.

- $\nabla \cdot \boldsymbol{f}$ differentiates the components of $\boldsymbol{f}$ and sums them, producing a scalar.
- $(\boldsymbol{f} \cdot \nabla)g$ differentiates the scalar field $g$ in the direction of $\boldsymbol{f}$, producing a scalar.
- $(\boldsymbol{f} \cdot \nabla)\boldsymbol{g}$ differentiates the components of $\boldsymbol{g}$ in the direction of $\boldsymbol{f}$, producing a vector.

### Curvilinear coordinates require scale factors

Do not use Cartesian gradient, divergence or curl formulas unchanged in curvilinear coordinates.

For orthogonal coordinates $(u, v, w)$, use the appropriate scale factors $h_u, h_v, h_w$ and the components of the vector field in the corresponding orthonormal basis.

In particular, the components $f_u, f_v, f_w$ in the standard orthogonal-coordinate formulas are components along the unit vectors $\boldsymbol{e}_u, \boldsymbol{e}_v, \boldsymbol{e}_w$.

### Differentiating in curvilinear coordinates

The scale factors account for the physical distance associated with a coordinate change:

$$ds_u = h_u\,du, \qquad ds_v = h_v\,dv, \qquad ds_w = h_w\,dw.$$

Consequently, the gradient contains factors $1/h_u$, $1/h_v$ and $1/h_w$. Divergence and curl have their own formulas and must not be obtained by simply inserting scale factors into the Cartesian expressions.

**Using vector identities without checking conventions**

Product rules and vector identities depend on the order of cross products and on which field the differential operator acts upon.

For example,

$$\nabla \cdot (\boldsymbol{f} \times \boldsymbol{g}) = \boldsymbol{g} \cdot (\nabla \times \boldsymbol{f}) - \boldsymbol{f} \cdot (\nabla \times \boldsymbol{g}).$$

If unsure, expand into Cartesian components and apply the ordinary product rule.

**Confusing the vector calculus product rules**

The product rules depend on whether the fields are scalar or vector-valued. In particular, distinguish

$$\nabla \cdot \boldsymbol{f} \qquad \text{from} \qquad \boldsymbol{f} \cdot \nabla.$$

The first is the divergence of a vector field and produces a scalar field. The second is a differential operator that differentiates whatever follows it in the direction of $\boldsymbol{f}$.

For scalar fields $\phi, \psi$ and vector fields $\boldsymbol{f}, \boldsymbol{g}$, remember:

- $\nabla(\phi\psi)$: use the scalar product rule.
- $\nabla \cdot (\phi\boldsymbol{f})$: use the divergence product rule.
- $\nabla \times (\phi\boldsymbol{f})$: use the curl product rule.
- $\nabla \cdot (\boldsymbol{f} \times \boldsymbol{g})$: use the cross-product divergence rule.
- $\nabla \times (\boldsymbol{f} \times \boldsymbol{g})$: use the cross-product curl rule.

Do not treat these as ordinary scalar multiplication rules: the order and type of each operation matter.

**Interpreting directional derivative terms**

In expressions such as

$$(\boldsymbol{f} \cdot \nabla)\boldsymbol{g},$$

the operator $\boldsymbol{f} \cdot \nabla$ acts on the vector field $\boldsymbol{g}$. In Cartesian coordinates, if

$$\boldsymbol{f} = (f_1, f_2, f_3), \qquad \boldsymbol{g} = (g_1, g_2, g_3),$$

then

$$(\boldsymbol{f} \cdot \nabla)\boldsymbol{g} = \begin{pmatrix} f_1 \dfrac{\partial g_1}{\partial x} + f_2 \dfrac{\partial g_1}{\partial y} + f_3 \dfrac{\partial g_1}{\partial z} \\ f_1 \dfrac{\partial g_2}{\partial x} + f_2 \dfrac{\partial g_2}{\partial y} + f_3 \dfrac{\partial g_2}{\partial z} \\ f_1 \dfrac{\partial g_3}{\partial x} + f_2 \dfrac{\partial g_3}{\partial y} + f_3 \dfrac{\partial g_3}{\partial z} \end{pmatrix}.$$

The result is a vector field. By contrast,

$$\nabla \cdot \boldsymbol{f} = \frac{\partial f_1}{\partial x} + \frac{\partial f_2}{\partial y} + \frac{\partial f_3}{\partial z}$$

is a scalar field. The dot in $\boldsymbol{f} \cdot \nabla$ combines the components of $\boldsymbol{f}$ with differential operators; it does not mean that the divergence is being calculated.

### 1.3.2 Methods

**Method 1.3.1** (Calculating the gradient in Cartesian coordinates)**.** For a scalar field $f(x, y, z)$, calculate the three partial derivatives:

$$\boxed{\nabla f = \frac{\partial f}{\partial x}\boldsymbol{e}_1 + \frac{\partial f}{\partial y}\boldsymbol{e}_2 + \frac{\partial f}{\partial z}\boldsymbol{e}_3}$$

(i) Differentiate $f$ with respect to $x$, holding $y, z$ constant.

(ii) Differentiate $f$ with respect to $y$, holding $x, z$ constant.

(iii) Differentiate $f$ with respect to $z$, holding $x, y$ constant.

(iv) Assemble the three components in the corresponding unit-vector directions.

**Method 1.3.2** (Calculating divergence in Cartesian coordinates)**.** For

$$\boldsymbol{f} = f_1\boldsymbol{e}_1 + f_2\boldsymbol{e}_2 + f_3\boldsymbol{e}_3,$$

use

$$\boxed{\nabla \cdot \boldsymbol{f} = \frac{\partial f_1}{\partial x} + \frac{\partial f_2}{\partial y} + \frac{\partial f_3}{\partial z}}$$

Differentiate each component with respect to its corresponding coordinate and add the results. The answer is a scalar.

**Method 1.3.3** (Calculating curl in Cartesian coordinates)**.** For

$$\boldsymbol{f} = f_1\boldsymbol{e}_1 + f_2\boldsymbol{e}_2 + f_3\boldsymbol{e}_3,$$

use the determinant mnemonic

$$\nabla \times \boldsymbol{f} = \begin{vmatrix} \boldsymbol{e}_1 & \boldsymbol{e}_2 & \boldsymbol{e}_3 \\ \partial_x & \partial_y & \partial_z \\ f_1 & f_2 & f_3 \end{vmatrix}.$$

Expand along the first row, remembering the alternating signs:

$$\boxed{\begin{aligned} \nabla \times \boldsymbol{f} = {} & (\partial_y f_3 - \partial_z f_2)\boldsymbol{e}_1 \\ & + (\partial_z f_1 - \partial_x f_3)\boldsymbol{e}_2 \\ & + (\partial_x f_2 - \partial_y f_1)\boldsymbol{e}_3. \end{aligned}}$$

The determinant is a mnemonic for the differential operator, rather than an ordinary determinant of numerical entries.

**Method 1.3.4** (Proving that the curl of a gradient is zero)**.** Let $f$ have continuous second partial derivatives. In Cartesian coordinates,

$$\nabla f = (f_x, f_y, f_z).$$

Then

$$\nabla \times (\nabla f) = \begin{pmatrix} f_{zy} - f_{yz} \\ f_{xz} - f_{zx} \\ f_{yx} - f_{xy} \end{pmatrix} = \mathbf{0},$$

since mixed partial derivatives are equal.

**Reusable principle:** Expand the curl and cancel the mixed partial derivatives in pairs.

**Method 1.3.5** (Proving that the divergence of a curl is zero)**.** Let $\boldsymbol{f} = (f_1, f_2, f_3)$ have continuous second partial derivatives. First write

$$\nabla \times \boldsymbol{f} = \begin{pmatrix} f_{3y} - f_{2z} \\ f_{1z} - f_{3x} \\ f_{2x} - f_{1y} \end{pmatrix}.$$

Taking the divergence gives

$$\begin{aligned} \nabla \cdot (\nabla \times \boldsymbol{f}) = f_{3yx} - f_{2zx} + f_{1zy} - f_{3xy} \\ + f_{2xz} - f_{1yz} = 0, \end{aligned}$$

by equality of mixed partial derivatives.

**Reusable principle:** Expand first, then pair equal mixed partial derivatives with opposite signs.

**Method 1.3.6** (Using the product rule for a gradient)**.** For scalar fields $f, g$, apply the ordinary product rule to each coordinate:

$$\boxed{\nabla(fg) = g\nabla f + f\nabla g.}$$

For example, the $x$-component is

$$\frac{\partial(fg)}{\partial x} = g\frac{\partial f}{\partial x} + f\frac{\partial g}{\partial x}.$$

The $y$- and $z$-components follow identically.

**Method 1.3.7** (Using the product rule for divergence)**.** For a scalar field $g$ and vector field $\boldsymbol{f}$,

$$\boxed{\nabla \cdot (g\boldsymbol{f}) = (\nabla g) \cdot \boldsymbol{f} + g(\nabla \cdot \boldsymbol{f})}$$

Write $g\boldsymbol{f} = (gf_1, gf_2, gf_3)$ and differentiate each component with respect to its corresponding coordinate. Apply the ordinary product rule and collect terms.

**Method 1.3.8** (Using the product rule for curl)**.** For a scalar field $g$ and vector field $\boldsymbol{f}$,

$$\boxed{\nabla \times (g\boldsymbol{f}) = (\nabla g) \times \boldsymbol{f} + g(\nabla \times \boldsymbol{f})}$$

Expand the curl componentwise and apply the ordinary product rule. The terms involving derivatives of $g$ assemble into $(\nabla g) \times \boldsymbol{f}$.

**Method 1.3.9** (Proving a vector identity)**.** When asked to prove a vector identity involving gradients, divergence, curl or products:

(i) Start with the more complicated side.

(ii) Expand using Cartesian components if the identity is unfamiliar.

(iii) Apply the ordinary product rule to every differentiated product.

(iv) Rearrange terms and use equality of mixed partial derivatives where applicable.

(v) Collect the resulting terms into the expression on the other side.

Alternatively, use a known vector identity, but check the order of all cross products and which field each derivative acts upon.

**Method 1.3.10** (Calculating the gradient in orthogonal curvilinear coordinates)**.** For orthogonal coordinates $(u, v, w)$ with scale factors $h_u, h_v, h_w$,

$$\boxed{\nabla f = \frac{1}{h_u}\frac{\partial f}{\partial u}\boldsymbol{e}_u + \frac{1}{h_v}\frac{\partial f}{\partial v}\boldsymbol{e}_v + \frac{1}{h_w}\frac{\partial f}{\partial w}\boldsymbol{e}_w}$$

(i) Identify the coordinate system and its scale factors.

(ii) Calculate the partial derivative with respect to each coordinate.

(iii) Divide each derivative by its corresponding scale factor.

(iv) Attach the appropriate unit vector to each component.

**Method 1.3.11** (Calculating divergence in orthogonal curvilinear coordinates)**.** For

$$\boldsymbol{f} = f_u\boldsymbol{e}_u + f_v\boldsymbol{e}_v + f_w\boldsymbol{e}_w,$$

use

$$\boxed{\begin{aligned}\nabla \cdot \boldsymbol{f} = \frac{1}{h_u h_v h_w}\Bigg[&\frac{\partial}{\partial u}(h_v h_w f_u)\\ &+\frac{\partial}{\partial v}(h_u h_w f_v)\\ &+\frac{\partial}{\partial w}(h_u h_v f_w)\Bigg].\end{aligned}}$$

**Exam trick:** Multiply each vector component by the two scale factors belonging to the other coordinates, differentiate with respect to its own coordinate, add the terms, and divide by $h_u h_v h_w$.

**Method 1.3.12** (Calculating curl in orthogonal curvilinear coordinates)**.** Use the scale-factor determinant:

$$\boxed{\nabla \times \boldsymbol{f} = \frac{1}{h_u h_v h_w}\begin{vmatrix} h_u\boldsymbol{e}_u & h_v\boldsymbol{e}_v & h_w\boldsymbol{e}_w \\ \partial_u & \partial_v & \partial_w \\ h_u f_u & h_v f_v & h_w f_w \end{vmatrix}.}$$

(i) Insert the correct scale factors and vector components.

(ii) Expand along the first row.

(iii) Apply each derivative to the entire expression beneath it.

(iv) Simplify each component and check the signs.

As with the Cartesian determinant, this is a mnemonic for a differential operator, not an ordinary numerical determinant.

**Method 1.3.13** (Checking a curvilinear-coordinate calculation)**.** After calculating a gradient, divergence or curl, check the following:

- The correct scale factors appear in each term.
- Derivatives act on all coordinate-dependent factors inside their brackets.
- Divergence is a scalar; gradient and curl are vectors.
- The result reduces to the Cartesian formula when all scale factors are 1 and the basis vectors are constant.

**Method 1.3.14** (Product rules for gradient, divergence and curl)**.** Let $\phi, \psi$ be scalar fields and $\boldsymbol{f}, \boldsymbol{g}$ be vector fields. Then

$$\begin{aligned}
\nabla(\phi\psi) &= \phi\nabla\psi + \psi\nabla\phi, \\
\nabla\cdot(\phi\boldsymbol{f}) &= \phi(\nabla\cdot\boldsymbol{f}) + \boldsymbol{f}\cdot\nabla\phi, \\
\nabla\times(\phi\boldsymbol{f}) &= \phi(\nabla\times\boldsymbol{f}) + \nabla\phi\times\boldsymbol{f}, \\
\nabla\cdot(\boldsymbol{f}\times\boldsymbol{g}) &= \boldsymbol{g}\cdot(\nabla\times\boldsymbol{f}) - \boldsymbol{f}\cdot(\nabla\times\boldsymbol{g}), \\
\nabla\times(\boldsymbol{f}\times\boldsymbol{g}) &= \boldsymbol{f}(\nabla\cdot\boldsymbol{g}) - \boldsymbol{g}(\nabla\cdot\boldsymbol{f}) \\
&\quad + (\boldsymbol{g}\cdot\nabla)\boldsymbol{f} - (\boldsymbol{f}\cdot\nabla)\boldsymbol{g}.
\end{aligned}$$

## 1.4 Integral Theorems

### 1.4.1 Mistakes and Tips

**Conservative Fields and Path Independence**

A conservative vector field has line integrals that depend only on the endpoints, not on the path taken between them. However, having zero curl is not automatically sufficient to establish that a field is conservative.

For a $C^1$ vector field on a simply connected open region of $\mathbb{R}^3$,

$$\nabla\times\boldsymbol{f} = \boldsymbol{0} \quad\Longrightarrow\quad \boldsymbol{f} \text{ is conservative.}$$

Without the simply connected condition, this implication can fail.

**Orientation in Stokes' Theorem**

The orientation of the boundary curve $C$ must be consistent with the chosen surface normal $\hat{\boldsymbol{n}}$. The positive orientation follows the right-hand rule: point your right thumb in the direction of $\hat{\boldsymbol{n}}$; your fingers curl in the positive direction around the boundary.

Equivalently, if $\hat{\boldsymbol{n}}_c$ is the outward unit co-normal to the boundary within the surface, then

$$\boxed{\hat{\boldsymbol{t}} = \hat{\boldsymbol{n}} \times \hat{\boldsymbol{n}}_c.}$$

Reversing either the surface normal or the boundary orientation changes the sign of the integral. Both orientations must be reversed together to preserve the value of the integral.

**The Divergence Theorem Requires a Closed Surface**

The divergence theorem relates the flux through a closed surface to the volume integral of the divergence inside it. If a surface is open, it may be necessary to introduce an additional surface to form a closed boundary, apply the divergence theorem, and subtract the flux through the additional surface.

**Direction of normal in Divergence Theorem**

The normal in divergence theorem must point outward from the enclosed volume. Using an inward normal reverses the sign of the flux.

**Stokes' Theorem Is Not Restricted to One Surface**

If several surfaces share the same boundary curve and have consistent orientations, the value of

$$\int_S (\nabla \times \boldsymbol{f}) \cdot \hat{\boldsymbol{n}}\, dS$$

is the same for each surface. This allows a complicated surface to be replaced by a simpler one, provided the boundary and orientation remain unchanged.

### 1.4.2 Methods

**Method 1.4.1** (Fundamental Theorem of Line Integrals)**.** Use this theorem whenever a vector field can be expressed as the gradient of a scalar potential.

If

$$\boldsymbol{f} = \nabla g$$

for some $C^1$ scalar field $g$, and $C$ is parametrised by $\boldsymbol{x}(t)$, $t \in [t_0, t_1]$, then

$$\boxed{\int_C \boldsymbol{f} \cdot d\boldsymbol{x} = g(\boldsymbol{x}(t_1)) - g(\boldsymbol{x}(t_0)).}$$

The key observation is the chain rule:
$$\begin{aligned}\frac{d}{dt} g(\boldsymbol{x}(t)) &= \nabla g(\boldsymbol{x}(t)) \cdot \boldsymbol{x}'(t) \\ &= \boldsymbol{f}(\boldsymbol{x}(t)) \cdot \boldsymbol{x}'(t).\end{aligned}$$

Integrating with respect to $t$ gives the result.

In practice:

(i) Identify the potential function $g$.

(ii) Determine the initial and terminal points.

(iii) Evaluate $g$ at both endpoints.

(iv) Subtract the initial value from the terminal value.

The path itself is irrelevant once the potential is known.

The curve may be piecewise $C^1$, provided the integral is well-defined.

**Method 1.4.2** (Testing Whether a Vector Field Is Conservative)**.** For a $C^1$ vector field

$$\boldsymbol{f} = (P, Q, R)$$

on a simply connected open region $V \subseteq \mathbb{R}^3$, use the following criterion:

$$\boxed{\boldsymbol{f} \text{ conservative} \iff \nabla \times \boldsymbol{f} = \mathbf{0}.}$$

Explicitly, calculate

$$\nabla \times \boldsymbol{f} = \begin{pmatrix} R_y - Q_z \\ P_z - R_x \\ Q_x - P_y \end{pmatrix}.$$

Thus, the conditions are

$$R_y = Q_z, \qquad P_z = R_x, \qquad Q_x = P_y.$$

If all three hold throughout the domain, the field is conservative, provided the domain is simply connected.

For a two-dimensional field

$$\boldsymbol{f} = (P, Q),$$

the corresponding condition on a simply connected open region of $\mathbb{R}^2$ is

$$\boxed{\frac{\partial Q}{\partial x} = \frac{\partial P}{\partial y}.}$$

If the domain is not simply connected, these conditions remain necessary but are not sufficient in general.

**Method 1.4.3** (Finding a Scalar Potential)**.** Suppose

$$\boldsymbol{f} = P(x, y, z)\boldsymbol{e}_1 + Q(x, y, z)\boldsymbol{e}_2 + R(x, y, z)\boldsymbol{e}_3$$

is conservative, so that

$$\nabla g = \boldsymbol{f}.$$

Then

$$\frac{\partial g}{\partial x} = P, \qquad \frac{\partial g}{\partial y} = Q, \qquad \frac{\partial g}{\partial z} = R.$$

To find $g$:

(i) Integrate the first component with respect to $x$:

$$g(x,y,z) = \int P(x,y,z)\,dx + \phi(y,z).$$

The integration term $\phi(y,z)$ must be retained because it may depend on the other variables.

(ii) Differentiate the resulting expression with respect to $y$, and compare with $Q$:

$$\frac{\partial g}{\partial y} = Q.$$

This determines $\phi(y,z)$ up to a function of $z$.

(iii) Differentiate with respect to $z$, and compare with $R$:

$$\frac{\partial g}{\partial z} = R.$$

This determines any remaining dependence on $z$.

(iv) Verify that all three partial derivatives reproduce the original vector field.

The potential is unique up to an additive constant on each connected component of the domain.

Once $g$ is obtained, line integrals can be evaluated using the fundamental theorem of line integrals.

**Method 1.4.4** (Using Path Independence)**.** If $\boldsymbol{f}$ is conservative, then for any two piecewise smooth curves $C_1, C_2$ with the same initial and terminal points,

$$\boxed{\int_{C_1} \boldsymbol{f}\cdot d\boldsymbol{x} = \int_{C_2} \boldsymbol{f}\cdot d\boldsymbol{x}.}$$

Equivalently, every closed curve has zero circulation:

$$\boxed{\oint_C \boldsymbol{f}\cdot d\boldsymbol{x} = 0.}$$

These statements are equivalent to conservativeness on a path-connected open region.

To exploit path independence, choose the simplest convenient path between the endpoints. For example, a piecewise linear path consisting of coordinate-parallel segments often simplifies the calculation.

Alternatively, if a potential is known, use the fundamental theorem of line integrals directly.

**Method 1.4.5** (Divergence Theorem)**.** Use the divergence theorem when calculating the flux of a vector field through a closed surface, particularly when the surface integral is difficult but the volume integral is simpler.

For a $C^1$ vector field $\boldsymbol{f}$ and a sufficiently smooth closed boundary surface $S = \partial V$,

$$\boxed{\oint_S \boldsymbol{f}\cdot\hat{\boldsymbol{n}}\,dS = \int_V \nabla\cdot\boldsymbol{f}\,dV.}$$

Here $\hat{\boldsymbol{n}}$ is the outward unit normal.

Procedure:

(i) Confirm that the surface is closed.

(ii) Identify the enclosed volume $V$.

(iii) Calculate the divergence:

$$\nabla \cdot \boldsymbol{f}.$$

(iv) Set up the volume integral over $V$, choosing an appropriate coordinate system.

(v) Evaluate the integral.

Useful coordinate forms include

$$dV = \begin{cases} dx\,dy\,dz & \text{Cartesian,} \\ r\,dr\,d\theta\,dz & \text{cylindrical,} \\ \rho^2 \sin\phi\,d\rho\,d\phi\,d\theta & \text{spherical.} \end{cases}$$

The divergence theorem is especially useful when the flux through individual faces or curved surfaces would be tedious to calculate separately.

**Method 1.4.6** (Closing an Open Surface for the Divergence Theorem)**.** If $S$ is an open surface, introduce an additional surface $S_c$ such that

$$S_{\text{closed}} = S \cup S_c$$

bounds a volume $V$.

Applying the divergence theorem gives

$$\oint_{S_{\text{closed}}} \boldsymbol{f} \cdot \hat{\boldsymbol{n}}\,dS = \int_V \nabla \cdot \boldsymbol{f}\,dV.$$

Splitting the surface integral,

$$\int_S \boldsymbol{f} \cdot \hat{\boldsymbol{n}}\,dS + \int_{S_c} \boldsymbol{f} \cdot \hat{\boldsymbol{n}}\,dS = \int_V \nabla \cdot \boldsymbol{f}\,dV.$$

Therefore,

$$\boxed{\int_S \boldsymbol{f} \cdot \hat{\boldsymbol{n}}\,dS = \int_V \nabla \cdot \boldsymbol{f}\,dV - \int_{S_c} \boldsymbol{f} \cdot \hat{\boldsymbol{n}}\,dS.}$$

Choose the closing surface so that its flux is easy to evaluate, ideally zero.

**Method 1.4.7** (Stokes' Theorem)**.** Use Stokes' theorem when calculating circulation around a closed curve, or when the line integral is difficult but the surface integral of the curl is simpler.

For a $C^1$ vector field $\boldsymbol{f}$, an oriented surface $S$, and its positively oriented boundary $C = \partial S$,

$$\boxed{\oint_C \boldsymbol{f} \cdot d\boldsymbol{x} = \int_S (\nabla \times \boldsymbol{f}) \cdot \hat{\boldsymbol{n}}\,dS.}$$

Procedure:

(i) Identify the boundary curve $C$.

(ii) Choose a convenient surface $S$ bounded by $C$.

(iii) Determine the corresponding surface normal using the right-hand rule.

(iv) Calculate the curl:
$$\nabla \times \boldsymbol{f}.$$

(v) Evaluate its flux through $S$.

An important simplification is that the surface can be replaced by any other suitably smooth surface with the same boundary, provided the orientation is consistent.

For example, a curved surface may be replaced by a planar disc if both have the same boundary curve.

**Method 1.4.8** (Green's Theorem)**.** Green's theorem is the two-dimensional analogue of Stokes' theorem.

For a $C^1$ vector field
$$\boldsymbol{f}(x, y) = P(x, y)\boldsymbol{e}_1 + Q(x, y)\boldsymbol{e}_2$$
and a positively oriented (anticlockwise) simple closed curve $C$ bounding a region $A$,
$$\boxed{\oint_C P\,dx + Q\,dy = \iint_A \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA.}$$

To apply it:

(i) Identify $P$ and $Q$.

(ii) Calculate
$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}.$$

(iii) Describe the enclosed region $A$.

(iv) Evaluate the resulting double integral.

The orientation is important: the standard formula assumes anticlockwise traversal. For clockwise traversal, the result changes sign.

Green's theorem can be viewed as Stokes' theorem applied to a planar surface in $\mathbb{R}^3$, with normal $\boldsymbol{e}_3$.

**Method 1.4.9** (Choosing Between the Integral Theorems)**.** When faced with a line, surface or volume integral, identify the geometric object and the operation being integrated.

- **Fundamental theorem of line integrals:** Use when integrating a gradient field along a curve. Replace the integral with the difference in potential at the endpoints.
- **Divergence theorem:** Use for flux through a closed surface. Replace the surface integral with a volume integral of divergence.
- **Stokes' theorem:** Use for circulation around a closed curve. Replace the line integral with a surface integral of curl.
- **Green's theorem:** Use for circulation around a closed planar curve. Replace the line integral with a double integral over the enclosed region.

The key relationships are

$$\int_C \nabla g \cdot d\boldsymbol{x} = g(B) - g(A),$$

$$\oint_{\partial V} \boldsymbol{f} \cdot \hat{\boldsymbol{n}}\, dS = \int_V \nabla \cdot \boldsymbol{f}\, dV,$$

$$\oint_{\partial S} \boldsymbol{f} \cdot d\boldsymbol{x} = \int_S (\nabla \times \boldsymbol{f}) \cdot \hat{\boldsymbol{n}}\, dS.$$

The general strategy is to replace a difficult integral with an equivalent integral over a higher- or lower-dimensional geometric object that is easier to evaluate.

**Method 1.4.10** (Useful Vector Calculus Identities)**.** The following identities are useful when simplifying vector fields and applying the integral theorems.

For sufficiently smooth scalar fields $\phi, \psi$ and vector fields $\boldsymbol{f}, \boldsymbol{g}$:

$$\begin{aligned}
\nabla \cdot (\nabla \times \boldsymbol{f}) &= 0, \\
\nabla \times (\nabla \phi) &= \boldsymbol{0}, \\
\nabla \cdot (\phi \boldsymbol{f}) &= \nabla \phi \cdot \boldsymbol{f} + \phi (\nabla \cdot \boldsymbol{f}), \\
\nabla \times (\phi \boldsymbol{f}) &= \nabla \phi \times \boldsymbol{f} + \phi (\nabla \times \boldsymbol{f}), \\
\nabla \cdot (\boldsymbol{f} \times \boldsymbol{g}) &= \boldsymbol{g} \cdot (\nabla \times \boldsymbol{f}) - \boldsymbol{f} \cdot (\nabla \times \boldsymbol{g}), \\
\nabla \times (\boldsymbol{f} \times \boldsymbol{g}) &= \boldsymbol{f}(\nabla \cdot \boldsymbol{g}) - \boldsymbol{g}(\nabla \cdot \boldsymbol{f}) \\
&\quad + (\boldsymbol{g} \cdot \nabla)\boldsymbol{f} - (\boldsymbol{f} \cdot \nabla)\boldsymbol{g}.
\end{aligned}$$

In particular,

$$\boxed{\nabla \cdot (\nabla \times \boldsymbol{f}) = 0}$$

and

$$\boxed{\nabla \times (\nabla \phi) = \boldsymbol{0}.}$$

These identities explain why the curl of a gradient field vanishes and why the divergence of a curl field vanishes.

# Chapter 2

# Algebra II

## 2.1 Rings, Fields and Subrings

### 2.1.1 Mistakes and Tips

**Determining unknown binary operations**

If not specified, assume that $R$ is a ring with $+$ and $\cdot$ by default

**Using universal conditions**

Work with any universal condition when manipulating or showing structure and then use strategically chosen elements

**Strategically choosing elements**

Use strategically chosen elements that exploit certain conditions such as $a+b$ or $ab$

**Dividing in rings**

Never divide by elements unless you know it is invertible. This is because rings do not require multiplicative inverses. Instead use addition / subtraction on either side and also distributivity.

**Make use of earlier results**

Utilise earlier results of the question where possible.

**Simplifying expansions**

Do not simplify expansions with hypotheses until it is the direct next step as you lose the algebra needed to show it is true.

**Utilising LHS and RHS**

Showing properties like associativity, commutativity and distributivity can be done easily by showing LHS = RHS.

**Efficiently forming Modular Arithmetic tables**

When dealing with integers modulo $n$, it is smart to use a multiplication and addition table. To quickly fill them in, consider how much over a multiple of $n$ your entry to the table is to find the residue.

**Using fields for subrings**

Using the fact that a subring of a ring is a ring itself allows you to quickly prove rings if you know that they are a subring of a larger field like $\mathbb{C}$.

**Dealing with subring questions**

When dealing with a subring question, make sure to use inherited elements and operations to show certain structures. For example 0 and 1.

**Showing subrings are the parent rings**

Showing subrings are the ring themselves requires showing $R \subset S$ as $S \subset R$ by default.

**Using counterexamples for disproofs**

Finding one failure allows you to disprove closure and the full ring quite easily. Use counterexamples strategically here.

### 2.1.2 Methods

**Method 2.1.1** (Determining a Ring)**.** To determine a ring, one should first check that the ring forms an abelian group with respect to the 'addition' operation and that the following requirements hold for the 'multiplication' operation:

i) **Existence of an Identity** - There exists an element $1 \in R$ such that $1 \cdot x = x \cdot 1 = x$, for all $x \in R$.

ii) **Associativity** - For any $x, y, z \in R$ we have $x(yz) = (xy)z$.

iii) **Distributivity** - For any $x, y, z \in R$ we have $x(y + z) = xy + xz$ and $(y + z)x = yx + zx$.

**Method 2.1.2** (Showing a Subring)**.** We can quickly show a subring by showing:

- the additive and multiplicative identities exist in the set,
- it's closed under addition
- it's closed under multiplication
- it contains additive inverses.

**Method 2.1.3** (Determining a Ring quickly)**.** Rather than displaying a set satisfies all of the properties of a ring, it is sufficient to show that the set is a subring of a ring. It is therefore a ring itself.

**Method 2.1.4** (Showing commutativity)**.** Showing commutativity requires introducing arbitrary elements and comparing expressions such as $(a+b)^2$ and the additive inverses to show equivalence.

**Method 2.1.5** (Quick way to see that a set is not a ring)**.** If multiplication is not closed by default then a set fails to be a ring as it is a requirement of the binary operation.

**Method 2.1.6** (Disproving $\mathbb{Z}/n$ as a field)**.** Showing that $\mathbb{Z}/n$ is not a field requires finding 2 single none zero elements that are zero divisors such that:

$$ab = 0 \bmod(n).$$

This immediately disproves a field. Check columns of a table here to do this quickly.

## 2.2 Arithmetic in $\mathbb{Z}$ and $F[x]$

### 2.2.1 Mistakes

**Defining the GCD**

Other than the algorithm, we need to actually categorise the greatest common divisor (GCD) by its properties. From this, we can produce a definition for the GCD

**Definition 2.2.1** (GCD via conditions)**.** The $\gcd(a,b)$ is the unique number $d \in \mathbb{N}_0$, such that the following two conditions hold:

i. $d$ is a common divisor of $a$ and $b$;

ii. $d$ is divisible by any common divisor of $a$ and $b$.

This number can be determined using the Euclidean Algorithm.

This number can also be defined according to the prime factorisation of any number:

**Definition 2.2.2** (GCD via formula)**.** Let $a, b \in \mathbb{N}$ have prime factorisations

$$a = p_1^{e_1} \cdots p_r^{e_r}, \quad b = q_1^{f_1} \cdots q_s^{f_s}.$$

Assume that $r \leq s$ and order the prime factors so that $p_i = q_i$ for $1 \leq i \leq k$, for some $0 \leq k \leq r$ and $p_i \neq q_j$ for all $i, j > k$ (note that $k = 0$ precisely when $p_i \neq q_j$ for all $i, j$ ). Show that

$$\gcd(a, b) = \prod_{i=1}^{k} p_i^{\min\{e_i, f_i\}}$$

(when $k = 0$ this is the "empty product" 1).

### Determining a common divisor

We know that $n$ is a common divisor of $a$ and $b$ if $n \mid \gcd(a, b)$ and vice versa

### GCD squared result

$$\gcd(a, b)^2 = \gcd\left(a^2, b^2\right).$$

### Coprime Meaning

Numbers are coprime if $\gcd(a, b) = 1$

### Bezout's Identity

Bézout's identity states that, for integers $a, b$, there exist $x, y \in \mathbb{Z}$ such that

$$\gcd(a, b) = ax + by.$$

In particular, if $\gcd(a, b) = 1$, then

$$ax + by = 1$$

for some integers $x, y$.

### Key Number Properties

**Basic divisibility properties**

- If $c \mid a$, then $c \mid ab$ for any $b \in \mathbb{Z}$.
- If $c \mid a$ and $c \mid b$, then $c \mid ma + nb$ for any $m, n \in \mathbb{Z}$.
- If $a \mid b$ and $b \mid c$, then $a \mid c$.
- If $a \mid b$ and $b \mid a$, then $a = \pm b$.
- If $a \mid b$ and $a \mid c$, then $a \mid (b \pm c)$.

- If $a \mid b$, then $a \mid (b + ka)$ for any $k \in \mathbb{Z}$.
- If $a \mid b$, then $a \mid -b$ and $-a \mid b$.

**Zero and units**

- If $a \neq 0$, then $a \mid 0$.
- For every $a \in \mathbb{Z}$, $1 \mid a$ and $-1 \mid a$.
- If $a \mid 1$, then $a = \pm 1$.
- $0 \mid a$ if and only if $a = 0$.
- If $a \mid b$, $a \neq 0$ and $b \neq 0$, then $|a| \leq |b|$.

**Products and powers**

- If $a \mid b$ and $c \mid d$, then $ac \mid bd$.
- If $a \mid b$, then $a^k \mid b^k$ for every integer $k \geq 1$.
- If $a \mid b$, then $a \mid bc$ for any $c \in \mathbb{Z}$.

**Cancellation and coprimality**

- If $ac \mid bc$ and $c \neq 0$, then $a \mid b$.
- If $a \mid bc$ and $\gcd(a, b) = 1$, then $a \mid c$.
- that If $\gcd(a, b) = 1$ and $m \mid a$ and $n \mid a$, for some integer $a$, then $mn \mid a$
- If $\gcd(a, b) = 1$ and $a \mid b$, then $a = \pm 1$.
- $\gcd(a, b) = 1$ if and only if there exist $m, n \in \mathbb{Z}$ such that $ma + nb = 1$.
- If $\gcd(a, b) = 1$, then $\gcd(a, bc) = \gcd(a, c)$.
- If $\gcd(a, b) = 1$, $a \mid c$ and $b \mid c$, then $|ab| \mid c$.

**Greatest common divisor (GCD)**

- $\gcd(a, b) \mid a$ and $\gcd(a, b) \mid b$.
- If $d \mid a$ and $d \mid b$, then $d \mid \gcd(a, b)$.
- $\gcd(a, b) = \gcd(b, a)$.
- $\gcd(a, b) = \gcd(|a|, |b|)$.
- $\gcd(a, 0) = |a|$.
- $\gcd(a, b) = \gcd(a, b - ka)$ for any $k \in \mathbb{Z}$.
- $\gcd(ka, kb) = |k| \gcd(a, b)$ for any $k \in \mathbb{Z}$.
- $\gcd\left(\frac{m}{\gcd(m,n)}, \frac{n}{\gcd(m,n)}\right) = 1$.
- Let $a, b, n \in \mathbb{N}$ and let $d$ be a common divisor of $a$ and $b$. Show that if $a \equiv b \pmod{n}$, then

$$\frac{a}{d} \equiv \frac{b}{d} \pmod{n/\gcd(d, n)}.$$

**The GCD of three or more integers**

We can define the gcd of three or more integers: If $a_1, \ldots, a_k \in \mathbb{Z}$, define $\gcd(a_1, \ldots, a_k)$ to be the largest integer $d$ such that $d \mid a_i$, for all $1 \leq i \leq k$.

**GCD and primes**

A common divisor cannot contain any prime power with an exponent greater than the exponent available in either number. The gcd takes the largest possible exponent for each prime, independently.

**Proving a gcd identity by comparing prime exponents**

Operations on gcd can sometimes be quicker with the prime characterisation

**What is what in Euclidean algorithm for polynomials**

When we do the Euclidean algorithm on a polynomial, the polynomial we are dividng by when we produce a 0 in the long division remainder is our *gcd*. In general, we have the remainder being the term we can't get rid of, the divisor being the polynomial 'waiting outside the bus stop' and the quotient being the polynomial 'on top of the bus stop'.

**Euclidean algorithm on polynomials in $\mathbb{Z}/n$**

When we do the Euclidean algorithm on a polynomial in $\mathbb{Z}/n$, we do normal division and then rewrite any numbers as the residue at each stage before continuing.

### 2.2.2 Methods

**Method 2.2.3** (Proving divisibility statmenets)**.** When proving divisiblity statement, its important to start from the definition $c \mid a \longleftrightarrow a = ck \quad$ for some $k \in \mathbb{Z}$. From here, we can use algebra to draw out what we need.

**Method 2.2.4** (Using the Euclidean Algorithm on integers)**.** Given positive integers $a > b$, repeatedly divide the previous divisor by the previous remainder and record the subsequent remainders:

$$\begin{aligned} a &= q_1 b + r_1, \\ b &= q_2 r_1 + r_2, \\ r_1 &= q_3 r_2 + r_3, \\ &\vdots \\ r_{k-2} &= q_k r_{k-1} + r_k, \\ r_{k-1} &= q_{k+1} r_k. \end{aligned}$$

The last nonzero remainder is

$$\gcd(a, b) = r_k.$$

**Method 2.2.5** (Finding the integers in the Bezout representation)**.**

1. Find the gcd using ordinary Euclidean divisions.
2. Rearrange the penultimate equation to express the gcd in terms of the preceding remainders.
3. Substitute expressions from earlier equations, working backwards.
4. Continue until the expression contains only the original $a$ and $b$.
5. Read off the coefficients $x, y$.

**Method 2.2.6** (Polynomial division over a field)**.** For polynomials $f, g \in F[x]$, where $F$ is a field and $g \neq 0$, there exist unique $q, r \in F[x]$ such that

$$f = qg + r, \quad r = 0 \text{ or } \deg r < \deg g.$$

Polynomial long division

1. Divide the leading term of the current dividend by the leading term of $g$.
2. Put the resulting term into the quotient.
3. Multiply it by $g$ and subtract.
4. Repeat until the remainder is zero or has degree smaller than $\deg g$.

**Method 2.2.7** (Euclidean Algorithm for polynomials)**.** For $f, g \in F[x]$, divide repeatedly:

$$\begin{aligned} f &= q_1 g + r_1, \\ g &= q_2 r_1 + r_2, \\ r_1 &= q_3 r_2 + r_3, \\ &\vdots \end{aligned}$$

Continue until the remainder is zero. The last nonzero remainder is a greatest common divisor, up to multiplication by a nonzero constant.

Two details matter in your questions:

- Monic gcd: Divide the final nonzero remainder by its leading coefficient so that its leading coefficient becomes 1.
- Bézout representation: Work backwards through the divisions to express the gcd as

$$\gcd(f, g) = A(x)f(x) + B(x)g(x).$$

## 2.3 Units, Zero divisors and integral domains

### 2.3.1 Mistakes and Tips

**Checking whether an element is a unit**

An element $a \in R$ is a unit if there exists $b \in R$ such that $ab = ba = 1$. Do not assume that every nonzero element is a unit unless $R$ is a field.

In $\mathbb{Z}/n\mathbb{Z}$, use

$$\overline{a} \text{ is a unit } \iff \gcd(a, n) = 1.$$

### Finding inverses modulo $n$

To find the inverse of $\overline{a}$ in $\mathbb{Z}/n\mathbb{Z}$, use the extended Euclidean algorithm to find

$$ax + ny = 1.$$

Reducing modulo $n$ gives $ax \equiv 1 \pmod{n}$, so

$$\overline{a}^{-1} = \overline{x}.$$

### Units form a group

The set of units $R^\times$ is a group under multiplication. In particular, products of units are units, and the inverse of a unit is a unit.

If $a, b \in R^\times$ with $ac = 1$ and $bd = 1$, then

$$(ab)(dc) = 1.$$

The order of the factors matters in a noncommutative ring.

### Distinguishing fields and integral domains

A field requires every nonzero element to be a unit, whereas an integral domain only requires that there are no nonzero zero divisors.

Every field is an integral domain, but an integral domain need not be a field. For example, $\mathbb{Z}$ is an integral domain but not a field.

### Finding zero divisors

To disprove that a ring is an integral domain, find $a, b \neq 0$ such that

$$ab = 0.$$

In $\mathbb{Z}/n\mathbb{Z}$, a nonzero residue $\overline{a}$ is a zero divisor precisely when

$$\gcd(a, n) > 1.$$

### Cancelling elements

Cancellation is valid in an integral domain:

$$ax = bx, \quad x \neq 0 \implies a = b.$$

In an arbitrary ring, cancellation may fail if $x$ is a zero divisor.

When proving cancellation, rearrange to obtain

$$(a - b)x = 0$$

and use the absence of nonzero zero divisors.

**Proving that a quotient ring is a field**

For $\mathbb{Z}/n\mathbb{Z}$, use the criterion

$$\mathbb{Z}/n\mathbb{Z} \text{ is a field } \iff n \text{ is prime.}$$

If $n$ is composite, write $n = ab$ with $1 < a, b < n$. Then $\overline{a}$ and $\overline{b}$ are nonzero but their product is zero.

**Prime versus irreducible**

A prime element satisfies

$$p \mid ab \implies p \mid a \text{ or } p \mid b.$$

An irreducible element is a nonzero nonunit whose every factorisation has a unit factor.

In an integral domain, every prime element is irreducible, but the converse does not hold in every integral domain.

**Checking whether two factors are genuinely different**

Two elements are associates if one is a unit multiple of the other. In a UFD, uniqueness is only up to order and associates.

Therefore, two different-looking factorizations do not disprove unique factorisation until all factors have been shown irreducible and the factors cannot be matched up to associates.

**Using degree in polynomial rings**

For nonzero polynomials over an integral domain,

$$\deg(fg) = \deg f + \deg g.$$

This is useful for proving that a polynomial is irreducible or that a polynomial ring has no zero divisors.

### 2.3.2 Methods

**Method 2.3.1** (Proving the unit criterion in $\mathbb{Z}/n\mathbb{Z}$)**.** To prove

$$\overline{a} \in (\mathbb{Z}/n\mathbb{Z})^\times \iff \gcd(a, n) = 1,$$

use the following two directions.

($\Leftarrow$) If $\gcd(a, n) = 1$, Bézout's identity gives integers $x, y$ such that

$$ax + ny = 1.$$

Hence $ax \equiv 1 \pmod{n}$, so $\overline{a}$ is a unit.

($\Rightarrow$) If $\overline{a}$ is a unit, then $ax \equiv 1 \pmod{n}$ for some integer $x$. Thus $ax + ny = 1$ for some integer $y$. Every common divisor of $a$ and $n$ divides 1, so $\gcd(a, n) = 1$.

**Method 2.3.2** (Proving that $\mathbb{Z}/n\mathbb{Z}$ is a field)**.** Use the fact that a commutative ring with identity is a field if and only if every nonzero element is a unit.

– If $n = p$ is prime, every nonzero $\overline{a}$ has $\gcd(a, p) = 1$. Therefore every nonzero element is

invertible.

– If $n$ is composite, write $n = ab$ with $1 < a, b < n$. Then

$$\overline{a} \neq \overline{0}, \qquad \overline{b} \neq \overline{0}, \qquad \overline{a}\,\overline{b} = \overline{0}.$$

Hence the ring is not a field.

**Method 2.3.3** (Finding a zero divisor in $\mathbb{Z}/n\mathbb{Z}$)**.** If $\gcd(a, n) = d > 1$, write

$$a = da', \qquad n = dn'.$$

Choose $b = n' = \frac{n}{d}$. Then $\overline{b} \neq \overline{0}$ and

$$ab = da'\frac{n}{d} = a'n \equiv 0 \pmod{n}.$$

Since $\overline{a} \neq \overline{0}$ when $n \nmid a$, both factors are nonzero, so $\overline{a}$ is a zero divisor.

**Method 2.3.4** (Proving cancellation in an integral domain)**.** Suppose $ax = bx$ with $x \neq 0$.

(i) Rearrange:
$$(a - b)x = 0.$$

(ii) Since $R$ has no nonzero zero divisors and $x \neq 0$, it follows that $a - b = 0$.

(iii) Therefore $a = b$.

This proves cancellation without dividing by $x$ or assuming that $x$ is a unit.

**Method 2.3.5** (Proving that the polynomial set of an integral domain is an integral domain)**.** Let $R$ be an integral domain and let $f, g \in R[x]$ be nonzero:

$$f = a_m x^m + \cdots, \qquad g = b_n x^n + \cdots,$$

where $a_m, b_n \neq 0$.

The leading coefficient of $fg$ is $a_m b_n$. Since $R$ has no nonzero zero divisors,

$$a_m b_n \neq 0.$$

Hence $fg \neq 0$. Thus $R[x]$ has no nonzero zero divisors. Since polynomial addition and multiplication inherit the relevant ring properties from $R$, the polynomial ring is an integral domain.

**Method 2.3.6** (Proving that a prime element is irreducible)**.** Let $p$ be prime in an integral domain $R$. By definition, $p$ is nonzero and not a unit.

Suppose

$$p = ab.$$

Since $p \mid ab$, primality gives $p \mid a$ or $p \mid b$.

Without loss of generality, suppose $p \mid a$. Write $a = pc$ for some $c \in R$. Then

$$p = ab = pcb.$$

Since $p \neq 0$ and $R$ is an integral domain, cancellation gives

$$1 = cb.$$

Hence $b$ is a unit. Therefore every factorisation of $p$ has a unit factor, so $p$ is irreducible.

**Method 2.3.7** (Showing that an element is irreducible)**.** To prove that a nonzero nonunit $a$ is irreducible, consider an arbitrary factorisation

$$a = bc.$$

Show that $b$ or $c$ must be a unit.

Useful techniques include:

- In $\mathbb{Z}$, use absolute values and the size of factors.
- In $F[x]$, use polynomial degrees:

$$\deg a = \deg b + \deg c.$$

- In other rings, use a suitable multiplicative norm or another invariant.

To disprove irreducibility, exhibit a factorisation into two nonunits.

**Method 2.3.8** (Disproving unique factorisation)**.** To show that an integral domain is not a UFD, find a nonzero nonunit with two genuinely different factorizations into irreducibles.

(i) Find two factorizations of the same element.

(ii) Prove every factor in both factorizations is irreducible.

(iii) Show that no reordering and multiplication by units makes the factorizations agree.

Merely finding two different-looking factorizations is insufficient.

## 2.4 Irreducibility in $F[x]$

### 2.4.1 Mistakes and Tips

**Irreducibility and the Degree of a Polynomial**

The relationship between roots and irreducibility depends on the degree of the polynomial.

- Degree 1: every non-constant linear polynomial over a field is irreducible.
- Degrees 2 and 3: a polynomial is irreducible if and only if it has no roots in the field.
- Degree 4: having no roots is not sufficient for irreducibility, since the polynomial may factorise into two irreducible quadratics.

More generally, if $f(x) \in F[x]$ is reducible, then

$$f(x) = g(x)h(x), \qquad 1 \leq \deg g, \deg h < \deg f.$$

In particular, a polynomial of degree $n$ is reducible if and only if it has a factor of degree at most $\lfloor n/2 \rfloor$.Thus, for degree 4, one must check for both linear factors and quadratic factors. A linear factor exists precisely when the polynomial has a root in $F$.

**Irreducibility Depends on the Field**

A polynomial can be irreducible over one field but reducible over a larger field.For example,

$$x^2 + 1$$

is irreducible in $\mathbb{R}[x]$ only if it has no real roots; however, it is reducible in $\mathbb{C}[x]$, since

$$x^2 + 1 = (x - i)(x + i).$$

Always identify the coefficient field when deciding irreducibility. The existence of a factorisation depends on which coefficients are permitted.

**Primitive Polynomials and Gauss's Lemma**

A polynomial in $\mathbb{Z}[x]$ is called primitive if the greatest common divisor of all its coefficients is 1.Gauss's lemma establishes the equivalence of irreducibility in $\mathbb{Z}[x]$ and $\mathbb{Q}[x]$ for primitive polynomials. For example,

$$2x + 2 = 2(x + 1)$$

is reducible in $\mathbb{Z}[x]$, but the distinction between integer and rational coefficients must be handled carefully: 2 is not a unit in $\mathbb{Z}$, whereas it is a unit in $\mathbb{Q}$. Therefore, do not apply Gauss's lemma directly to a polynomial whose coefficients have a non-trivial common divisor. First extract the content (the GCD of its coefficients).

**The Rational Root Test Gives Necessary Conditions**

If $f(x) \in \mathbb{Z}[x]$ has a rational root $p/q$, written in lowest terms, then

$$p \mid a_0, \qquad q \mid a_n.$$

These conditions restrict the possible rational roots but do not guarantee that any candidate is actually a root.A candidate must still be substituted into $f(x)$ and checked. Furthermore, having no rational roots proves irreducibility over $\mathbb{Q}$ only when the degree is 2 or 3. For degree 4 or higher, the polynomial may factorise into higher-degree factors without possessing rational roots.

**Eisenstein's Criterion Is Sufficient, Not Necessary**

Eisenstein's criterion provides a sufficient condition for irreducibility over $\mathbb{Q}$, not a necessary one. A polynomial failing the criterion for a particular prime is not necessarily reducible. For example, a polynomial may be irreducible even though no prime satisfies the required divisibility conditions in its original form. One should also consider translating the polynomial before applying the criterion, since a substitution such as $x \mapsto x + 1$ can produce a polynomial satisfying Eisenstein's criterion.

### 2.4.2 Methods

**Method 2.4.1** (Testing Irreducibility Using Polynomial Degree)**.** To test whether $f(x) \in F[x]$ is irreducible, use its degree to determine which factorisations are possible.

- Degree 1: automatically irreducible.
- Degree 2 or 3: test whether $f(x)$ has a root in $F$. If it has none, it is irreducible.
- Degree 4: first test for roots in $F$, then test whether it factorises into two quadratics.

More generally, for a polynomial of degree $n$, it is sufficient to rule out factors of degrees

$$1, 2, \ldots, \left\lfloor \frac{n}{2} \right\rfloor .$$

This follows because if $f = gh$ is a non-trivial factorisation, at least one of $g, h$ must have degree at most $n/2$.

**Method 2.4.2** (Rational Root Test)**.** For

$$f(x) = a_0 + a_1 x + \cdots + a_n x^n \in \mathbb{Z}[x],$$

suppose $p/q \in \mathbb{Q}$ is a root, where

$$\gcd(p, q) = 1, \qquad q > 0.$$

Then

$$\boxed{p \mid a_0, \qquad q \mid a_n.}$$

To apply the test:

(i) List the integer divisors of the constant coefficient $a_0$.

(ii) List the positive divisors of the leading coefficient $a_n$.

(iii) Form all possible fractions $p/q$, including negative numerators.

(iv) Substitute the candidates into $f(x)$ to identify any actual roots.

If $a_0 = 0$, then $x$ is already a factor and should be extracted before applying the test to the remaining polynomial.

If $f$ has degree 2 or 3, finding no rational roots proves that it is irreducible over $\mathbb{Q}$.

**Method 2.4.3** (Gauss's Lemma: Reducing Irreducibility to Integer Coefficients)**.** For a non-zero polynomial $f \in \mathbb{Z}[x]$, define its content by

$$\mathrm{cont}(f) = \gcd(a_0, a_1, \ldots, a_n),$$

taken to be positive.

A polynomial is primitive precisely when its content is 1.

Gauss's lemma states that if $f \in \mathbb{Z}[x]$ is primitive and non-constant, then

$$\boxed{f \text{ irreducible in } \mathbb{Z}[x] \iff f \text{ irreducible in } \mathbb{Q}[x].}$$

A useful consequence is that a primitive polynomial reducible over $\mathbb{Q}$ must also admit a factorisation into non-constant polynomials in $\mathbb{Z}[x]$.

To test irreducibility in $\mathbb{Q}[x]$:

(i) Calculate the GCD of the coefficients.

(ii) If necessary, extract the content to obtain a primitive polynomial.

(iii) Work with the primitive polynomial in $\mathbb{Z}[x]$.

(iv) Apply integer factorisation techniques, such as the rational root test or Eisenstein's criterion.

If the original polynomial is non-primitive, its non-unit integer content already gives a non-trivial factorisation in $\mathbb{Z}[x]$, provided the content has absolute value greater than 1.

**Method 2.4.4** (Irreducibility Preserved by Multiplication by a Unit)**.** Let $R$ be a commutative ring, let $f \in R$ be irreducible and let $u \in R^\times$ be a unit.

Then

$$\boxed{f \text{ irreducible} \implies uf \text{ irreducible.}}$$

Indeed, suppose

$$uf = ab.$$

Multiplying by $u^{-1}$ gives

$$f = (u^{-1}a)b.$$

Since $f$ is irreducible, either $u^{-1}a$ or $b$ is a unit. Hence either $a$ or $b$ is a unit.

Thus multiplication by a unit does not change irreducibility.

This is particularly useful when working over $\mathbb{Q}[x]$, where every non-zero rational constant is a unit, and when extracting non-zero constant factors from polynomials.

**Method 2.4.5** (Eisenstein's Criterion)**.** Let

$$f(x) = a_0 + a_1x + \cdots + a_nx^n \in \mathbb{Z}[x], \qquad n \geq 1.$$

If there exists a prime $p$ such that

$$\begin{aligned} p &\mid a_0, a_1, \ldots, a_{n-1}, \\ p &\nmid a_n, \\ p^2 &\nmid a_0, \end{aligned}$$

then

$$\boxed{f(x) \text{ is irreducible in } \mathbb{Q}[x].}$$

To apply the criterion:

(i) Identify a prime $p$ dividing every coefficient except the leading coefficient.

(ii) Check that $p$ does not divide the leading coefficient.

(iii) Check that $p^2$ does not divide the constant coefficient.

(iv) Conclude irreducibility over $\mathbb{Q}$.

The criterion also implies that $f$ is primitive: since $p \nmid a_n$, no prime dividing all coefficients can be $p$, and since every other coefficient is divisible by $p$, any common divisor must divide $a_n$ as well as the other coefficients. The conditions therefore force the content to be 1.

**Method 2.4.6** (Applying Eisenstein After a Change of Variable)**.** Sometimes a polynomial does not satisfy Eisenstein's criterion directly, but a translated polynomial does.

Consider

$$g(x) = f(x + c), \qquad c \in \mathbb{Z}.$$

If $g(x)$ satisfies Eisenstein's criterion for some prime $p$, then $g(x)$ is irreducible in $\mathbb{Q}[x]$.

Since the substitution $x \mapsto x + c$ is invertible, with inverse $x \mapsto x - c$, irreducibility is preserved:

$$\boxed{f(x) \text{ irreducible } \iff f(x + c) \text{ irreducible.}}$$

Thus, when direct application fails, try a small integer translation and expand the resulting polynomial before checking the divisibility conditions.

More generally, substitutions $x \mapsto x + c$ for $c \in \mathbb{Z}$ preserve irreducibility over $\mathbb{Q}$.